Sibeen Kim

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Research Goal

I want robots that perform physical tasks humans do. My research **transfers human prior into data that robots can learn from**. Previously, I converted human motion, from motion capture to in-the-wild videos, into robot data. Currently, I am using agents that possess human prior to generate robot data.

Education

Korea Advanced Institute of Science and Technology (KAIST) Mar 2025 -
M.S./Ph.D. in Artificial Intelligence (Advisor: Jaegul Choo)

Korea University Mar 2018 - Feb 2025
B.S. in Biomedical Engineering, GPA: 4.00 / 4.5

Gyeonggi Science High School Mar 2015 - Feb 2018
School for Gifted Students in Science

Ongoing Project

Oct 2026 - **Agentic Skill Composition for Functional Tasks in Dexterous Manipulation**
***Sibeen Kim**, Jaegul Choo, Minho Park.*
TL;DR: We use agents with human prior to generate robot data for compositional, functional manipulation tasks.

Oct 2026 - **Agentic Data Generation for Variational Automation in Dexterous Manipulation**
Sibeen Kim, Kyungmin Lee*, Dongyoon Hwang*, Jaegul Choo, Hojoon Lee.*
TL;DR: We use agents with human prior to generate robot data for varying instances of a manipulation task.

Publication

arXiv'26 **FlashDexRetarget: Accelerating Dexterous Manipulation Data Generation through Multi-Motion Retargeting**
Kyungmin Lee, **Sibeen Kim***, Dongyoon Hwang*, Yoonsang Oh*, Donghu Kim, Youngdo Lee, I Made Aswin Nahrendra, Jaegul Choo, Hojoon Lee.*
TL;DR: We transfer human hand-object motion into physically grounded robot manipulation data.
Website Paper Code

CoRL'26 (spotlight) **PHUMA: Physically Reliable Humanoid Locomotion Dataset**
Kyungmin Lee, **Sibeen Kim***, Youngdo Lee, Minho Park, Hyunseung Kim, Dongyoon Hwang, Donghu Kim, Hojoon Lee, Jaegul Choo.*
TL;DR: We transfer human whole-body motion into physically reliable robot locomotion data.
Website Paper Code **(200+☆)**